

Tom George

Post-doctoral fellow in ML and theoretical neuroscience,
also building open-source software for science
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ACADEMIC POSITIONS	Post-doctoral Fellow and Canadian Neuroanalytic Scholar, Mila (Quebec AI Institute) Advisors: Prof. Blake Richards (McGill), Prof. Guillaume Lajoie (UdeM)	2025-
EDUCATION	PhD, Sainsbury Wellcome Centre, UCL Advisors: Caswell Barry (UCL), Claudia Clopath (Imperial), Kimberly Stachenfeld (DeepMind) and Caswell Barry (UCL). Member of Google DeepMind NeuroLab . Herchel Smith Scholarship, Harvard University Prestigious and fully funded scholarship to study at Harvard. Advisors: Prof. Cengiz Pehlevan & Prof. Sam Gershman BA, MA & MSci, Physics (Natural Sciences), University of Cambridge 1 st Class (Part III), 1 st Class (Part II), 1 st Class (Part I)	2020–25 2019–20 2015–19
AWARDS	Canadian Neuroanalytic Scholarship Post-doctoral fellowship of 140,000CAD UCL Early Career Neuroscience Prize Junior category, £500 Herchel Smith Scholarship \$80000 fully-funded scholarship to Harvard Emmanuel College Cambridge John Mainhood Prize for high academic achievement	2026–28 2023 2019 2016–18
SOFTWARE	RatInABox Creator and maintainer of a popular python package for generation of motion and neural data in spatial environments (>75,000 PyPI downloads). Link . SIMPL Open-source package for neural latent data analysis. Link .	2022 2024
INDUSTRY	X AI Residency, Google Foundation models for geophysics. Mountain View, California. ML consultant, Mitpl ML tools for automated business valuations	2025 2023–24
TECHNICAL SKILLS	Programming Python (very proficient inc. Jax), bash, Julia [github.com/TomGeorge1234]. Machine Learning NLPs, ANNs, CNNs, RNNs, generative models (PyTorch, tensorflow). Languages English (Native), Spanish (DELE B1 equivalent).	
TEACHING & OUTREACH	TReND-CaMinA Founder and organizer of a neuroscience and ML summer school in Africa. Secured \$300,000 funding. (Ghana 2023, Rwanda 2024, Zambia 2025, Kenya 2026) NeuroAI summer school University of Amsterdam, invited lecturer on model-free RL.	2023–26 2024
ACADEMIC PLACEMENTS	Caltech Summer Undergraduate Research Fellow, Prof. A. Thompson Okinawa Computational Neuroscience Course (OCNC) Summer school participant Okinawa Institute of Science and Technology (OIST) Research intern with Prof. Tomoki Fukai	2018 2022 2022
SELECTED INVITED TALKS	Mila, Quebec AI Institute SIMPL; fast latent variable analysis for neural data Oxford University, Cortex Club RatInABox (see open source). Google DeepMind, London Optimizing internal representations (NeuroLab workshop) Imperial, Pint of Science Public outreach talk on “The language of the brain” ICLR TinyPapers workshop oral Neural oscillations and eligibility traces, Kigali, Rwanda Google DeepMind, London The Helmholtz hippocampus (NeuroLab workshop) UCL Neuroscience Symposium (prize winner) How hippocampus learns predictive maps? Spring Hippocampal Research Conference, Verona RatInABox (see open source). Mila Neural-AI reading group Biological Reinforcement Learning in the Hippocampus.	2025 2024 2023 2022
SELECTED PUBLICATIONS	T. M. George (2025). SIMPL: Scalable and hassle-free optimisation of neural representations from behaviour. <i>International Conference on Learning Representations (ICLR)</i> T. M. George (2023). A generative model of the hippocampal formation trained with theta driven local learning rules. <i>Advances in Neural Information Processing Systems (NeurIPS)</i> E. Thompson, ..., T. M. George et al. (2024) Replay of procedural experience is independent of the hippocampus. (<i>Nature</i>)	

T. M. George (2023). Theta sequences as eligibility traces: A biological solution to credit assignment. *Tiny Papers Track at ICLR 2023, Kigali, Rwanda*. [Paper](#) (accepted for oral pres.)

T. M. George, M. Rastogi, W. de Cothi, C. Clopath, K. Stachenfeld, & C. Barry (2024). RatInABox: An open source toolkit for modelling locomotion and neuronal activity in continuous environments. [eLife](#)

T. M. George*, W. de Cothi*, K. Stachenfeld, & C. Barry (2022). Rapid learning of predictive maps with STDP and theta phase precession. [eLife](#).

T. M. George, G. E. Manucharyan, & A. F. Thompson (2021). Deep learning to infer eddy heat fluxes from sea surface height patterns of mesoscale turbulence. [Nature Communications](#).

T. M. George & P. Liò (2019). Unsupervised Machine Learning for Data Encoding applied to Ovarian Cancer Transcriptomes. [bioRxiv](#).

REVIEWING **Area chair, ICLR 2023 and 2024** | TinyPapers workshop
Reviewer, NeurIPS 2023 | Associative Memory & Hopfield Networks workshop

CONFERENCE POSTERS ETC. COSYNE 2024 poster: C. Barry, M. Rastogi, W. de Cothi, C. Clopath, K. Stachenfeld, **T. M. George** (2024) RatInABox: A unified Python framework for modelling spatial behaviour and neural data

COSYNE 2024 poster: **T. M. George**, C. Barry, K. Stachenfeld, C. Clopath, T. Fukai (2024) The Helmholtz Hippocampus: A biologically plausible generative model of the Hippocampal formation

NeurIPS 2023 poster: **T. M. George**, C. Barry, K. Stachenfeld, C. Clopath, T. Fukai (2024) The Helmholtz Hippocampus: A biologically plausible generative model of the Hippocampal formation

COSYNE 2024 poster: **T. M. George**, W. de Cothi, K. Stachenfeld, C. Barry (2022). Rapid learning of predictive maps with STDP and theta phase precession.

SFN 2022 poster & Cosyne 2023 Talk: E. Thompson, L. Rollik, **T. M. George*** et al., Replay of motor sequences in DLS during consolidation using an unsupervised point process model.

SFN 2021 poster: A. Onih, **T. M. George**, S. Nierwetberg & A. Akrami. Pupil dilation as a proxy for statistical learning in freely moving mice and humans.

Reservoir networks for unsupervised statistical learning (2021) (Github <https://tinyurl.com/3mmpij6r>). Advisors: Dr. Athena Akrami & Prof. Claudia Clopath

Pupillometry protocol and pipeline for studying temporal structure learning in humans (2021) (Github, <https://tinyurl.com/2nr4c72t>). Advisors: Dr. Athena Akrami & Prof. Claudia Clopath

Deep learning to explain mixed selectivity of neurons in the prefrontal cortex (2020) (<https://tinyurl.com/2p8zdd8w>). Advisor: Prof. Cengiz Pehlevan & Prof. Sam Gershman

REFEREES **Prof. Blake Richards** (postdoc advisor), McGill/Mila | blake.richards@mila.quebec
Prof. Caswell Barry (PhD supervisor), UCL | caswell.barry@ucl.ac.uk
Prof. Kimberly Stachenfeld, DeepMind/Columbia | stachenfeld@deepmind.com